Human Review: Schlumprecht Szlenk-radius converse

Original automatic proof: `solution_packet.pdf`
Companion notation rewrite: `human_requested_better_notation.pdf`
Checked date: 14 June 2026
Status: Manually checked.
Verdict: Appears correct.

Human Comments

The proposed proof appears to be correct. It gives the reverse inequality for the Szlenk-radius formula in Schlumprecht's space, and therefore appears to complete the converse left open in the source paper.

The idea of the proof is to show that, if a vector has norm smaller than the proposed radius, then, after harmlessly reducing to the finitely supported case, one can perturb it far out by

$$\pm\alpha e_N,$$

where $\alpha > \varepsilon/2$. These two perturbations are

$$x + \alpha e_N \quad \text{and} \quad x - \alpha e_N.$$

They are weakly close to x, because $e_N \to 0$ weakly, but they are separated by

$$\|(x + \alpha e_N) - (x - \alpha e_N)\|_{\mathcal{S}} = 2\alpha > \varepsilon.$$

Thus they witness diameter $> \varepsilon$ in every weak neighbourhood of x, provided they remain in the unit ball.

The key technical step is exactly this unit-ball control:

$$\|x \pm \alpha e_N\|_{\mathcal{S}} \leq 1$$

whenever

$$\|x\|_{\mathcal{S}} < a_\alpha, \qquad a_\alpha = \inf_{n\geq 1} \frac{\varphi(n+1) - \alpha}{\varphi(n)}.$$

This is the part where one has to control the size of the new vectors in the recursive Schlumprecht norm. The scalar majorant M_α, the logarithmic product lemma, and the local comparison argument are all used to prove precisely this point.

I checked the main scalar estimate and the comparison argument. The argument appears sound. The original notation was somewhat difficult to follow, so I requested a separate improved-notation rewrite of the proof. That rewrite is included only as a human-requested notational aid; it is not the original automatic output.

Short Version

The proof works by taking x below the proposed radius and perturbing it far out by $\pm\alpha e_N$. The perturbations are weakly close to x and are more than ε apart. The main work is to show that both perturbed vectors remain in the unit ball of Schlumprecht's space. Once this is proved, the reverse Szlenk-radius inclusion follows by the standard density and weak-closedness argument.

Literature and Follow-Up

This appears to be a serious candidate full solution to the converse problem in the source paper. I recommend checking the argument with the authors, both to confirm correctness and to understand whether the result or method was already known to them.

Review Outcome

Archive this packet as an apparently correct candidate full solution. The main mathematical content is the far-coordinate perturbation argument and the scalar estimate controlling the Schlumprecht norm after adding $\pm\alpha e_N$.